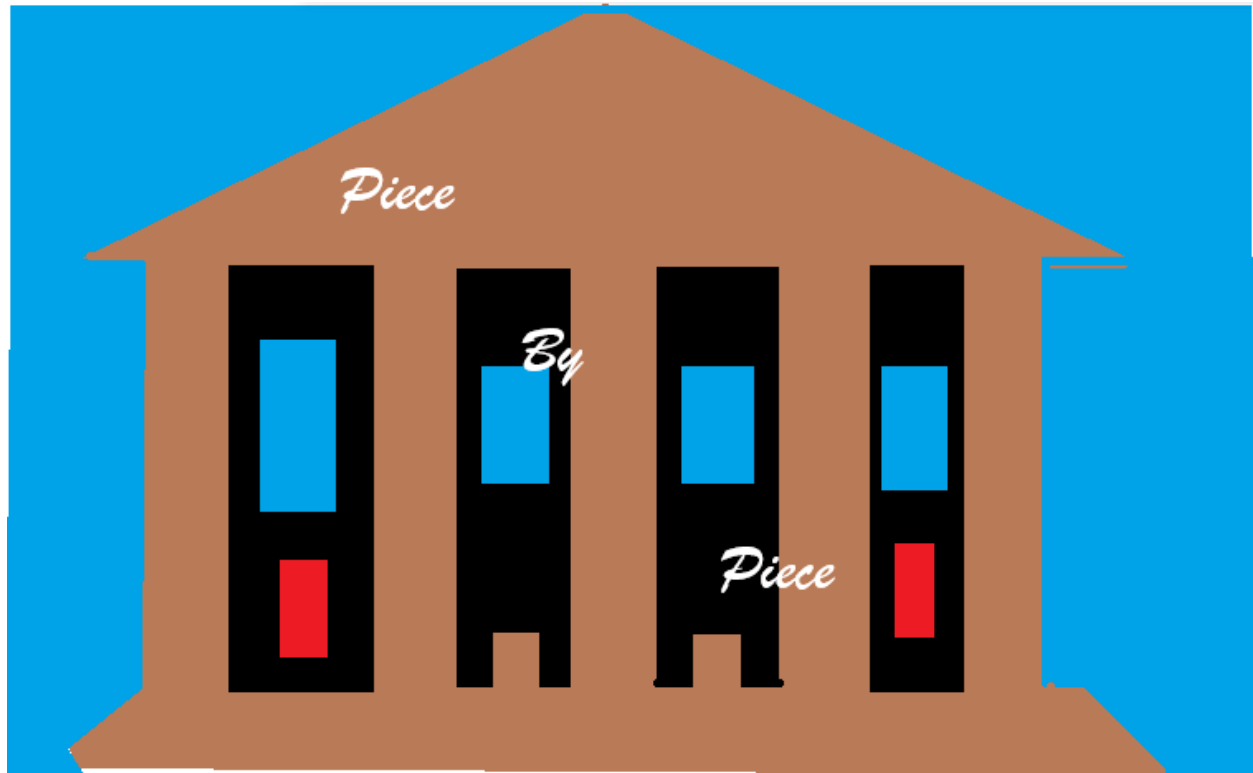


Piece by Piece

By Zachary Sansone



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1 Game Design Document

1.1 Overview

Piece by Piece will be a 2D incremental museum management game in which the player recovers artifacts from archaeological sites and restores them to display in their museum. The player will send a recovery team to these sites to search for fragments of artifacts. The recovery team can also sometimes find things that will be automatically sold and give the player a small amount of funds for their museum. These funds can be used to increase the speed of the teams and the chance of finding fragments.

Once found, fragments are placed on the artifact's designated display case. In this form, artifact fragments generate a very small amount of funds per second. The player can also click to clean and touch up each fragment piece. Cleaned fragment pieces generate marginally more funds. Once every fragment has been collected, the player can choose to assemble the artifact, which generates much more funds per second and gives a one-time increase to reputation points, which unlock new upgrades.

This version of the game will include two archaeological sites for the player to send their team to. The first area will contain an artifact that will not be accessible to the player until they have made progress in the second area through the purchase of a reputation-locked tool. The player wins the game by restoring and displaying all artifacts in the museum.

Piece by Piece will combine a simplified version of the generation and upgrade progress of **Cookie Clicker** (Orteil, 2013, Figure 1) with the collection and excavation themes of **Digseum** (Rat Monthly, 2024, Figure 2, 3). This game will use a pixel-art art style.

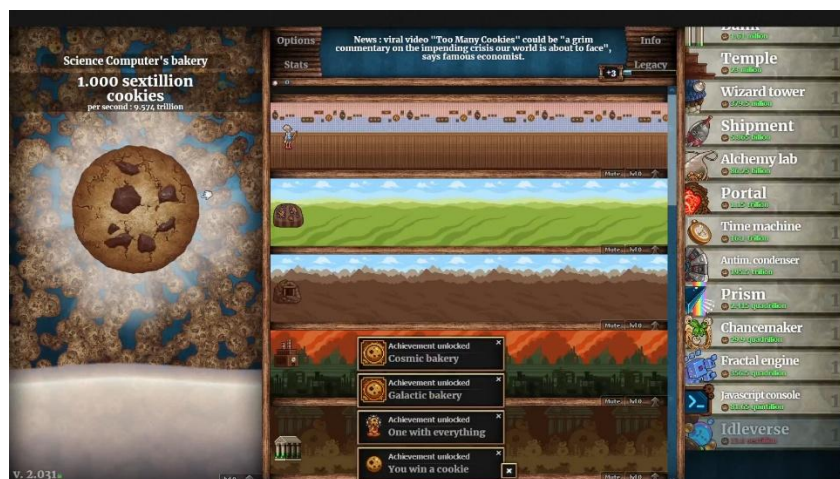


Figure 1. Cookie Clicker from [steam page](#)

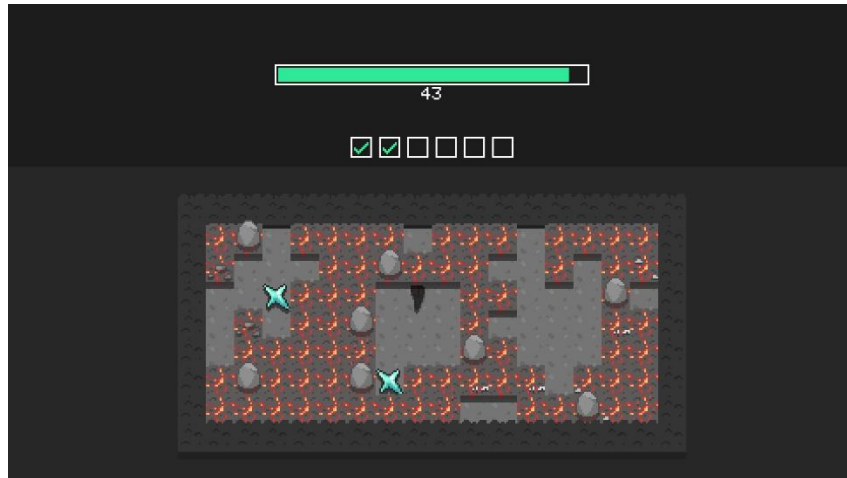


Figure 2. Digseum collection from [steam page](#)

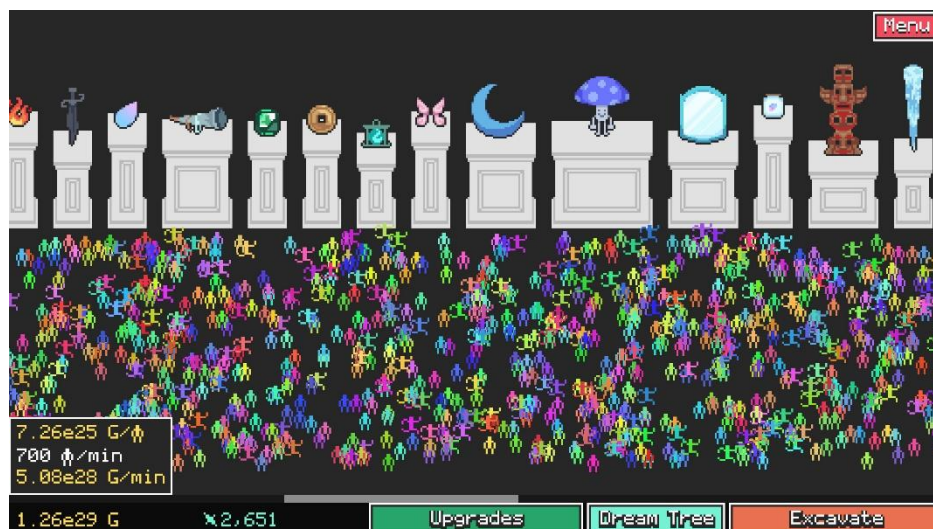


Figure 3. Digseum display from [steam page](#)

1.2 Game Mechanics and Features

1.2.1 Interactions, Rules, and resources

1.2.1.1 Recovery Team and Fragment Excavation

The player manages a recovery team that can be assigned to any unlocked archeological site. The selected site attempts to extract an artifact fragment at regular intervals.

At the beginning of the game, the excavation results in success or failure every 10 seconds. At the beginning of the game, the chance of successfully extracting a fragment depends on the selected artifact's multiplier and the area's base success chance. On failure, the player will receive a small amount of funds within a range of values unique to each area. The

player can spend funds to reduce the time between attempts and increase the chance of a successful attempt.

The player selects the active artifact from the chosen site's information panel. The recovery team continues attempting to excavate fragments for that artifact until the player selects a different valid artifact or all of its fragments have been recovered.

Certain artifacts cannot be selected by the player until the player has purchased an upgrade to unlock a specific tool that lets the recovery team attempt to extract that artifact's fragments.

The player can manually switch the recovery team to another unlocked area at any time, pausing the excavation timer for the previous area.

Once all artifacts for an area are unlocked, the player receives a boost in income for that specific exhibit.

1.2.1.2 Exhibit Screens and Fragment collection

In this version of the game, the museum will contain two exhibit screens, themed for each of the archaeological sites. Each exhibit contains initially empty display pedestals, cases, frames, or whatever else is appropriate for each artifact that is available for that exhibit.

Extracted artifact fragments are automatically delivered to a case next to the display which has a fixed number of spaces. Empty spaces for un-extracted fragments and are filled in when delivered. (Figure 4)

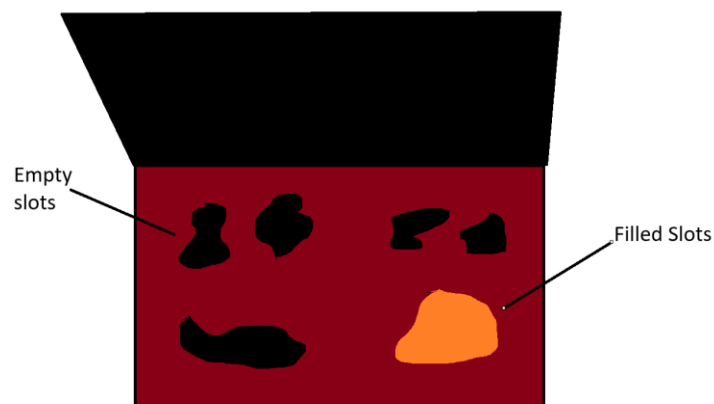


Figure 4. Case Example

1.2.1.3 Fragment cleaning

When fragments are extracted by the recovery team, they arrive uncleaned. Uncleaned fragments in their case generate a very small amount of passive income which can be seen

by hovering over the case as a total for all fragments currently inside the case. When the player selects the case, they can choose fragments to clean. By clicking and dragging their mouse over the fragment, the fragment changes color to a brighter, cleaner version. (Figure 5) Once a sufficient amount of the fragment has been cleaned off, the remaining dirt the player may have missed gets cleaned automatically and the fragment is registered as cleaned.

Cleaned fragments generate an increased income. To fully assemble an artifact, the player must clean all fragments.

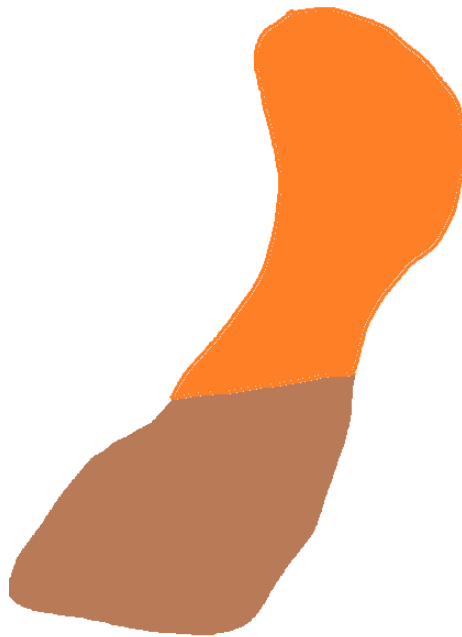


Figure 5. Cleaning

1.2.1.4 Artifact Assembly

Once every fragment for an artifact has been recovered and cleaned, an assemble artifact button becomes available within the case. An exclamation mark hovers above the case on the exhibit screen and menu UI to show players when they have acquired all fragments of an artifact, regardless of whether they are cleaned yet or not.

Once the player selects the assemble button, a restoration animation plays, and the fragments float out of their case towards the display, rotate, and move into the correct positions. The fragments snap together with some sound and visual effects and gently drop the completed artifact into place. (Figure 6)

A complete artifact generates much more income than when fragmented and gives the player a one-time reputation reward.

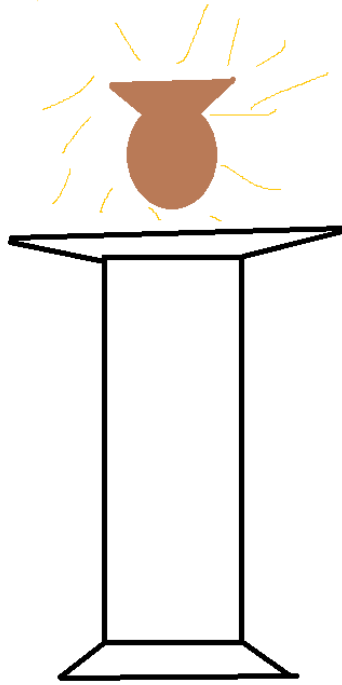


Figure 6. Artifact Display

1.2.1.5 Museum Funding

Museum Funds are the primary spending resource of the game. The player earns funds from a failed fragment-recovery attempt and from passive gains from museum visitors for fragments and completed artifacts present in the museum.

Uncleaned fragments generate the smallest amount, cleaned fragments generate marginally more, and completed artifacts generate the most. Museum funding can be used to purchase upgrades for the recovery team and excavation tools.

1.2.1.6 Upgrades

The player can spend museum funding on upgrades to improve the recovery team. Each area has its own set of upgrades and an initial value for each upgradable statistic.

The current planned upgrades are to the speed of excavation and the chance that fragments are discovered.

Each upgrade becomes more expensive after it is purchased.

1.2.1.7 Reputation

Reputation represents how popular the museum is. The player earns reputation when an artifact is fully completed and displayed in its exhibit. Reputation is not spent but rather used as a requirement to unlock certain upgrades and archaeological sites.

1.2.1.8 Excavation Tools

Some artifacts cannot be excavated with the recovery team's current equipment and require a specific excavation tool.

This tool is unlocked after the player reaches a certain amount of reputation and pays a cost in funds. Once unlocked, the recovery team can extract any fragments that that tool is made to extract.

1.2.1.9 Archaeological Sites and Progression

This version of the game will contain two archaeological sites and six total artifacts.

The first exhibit contains three artifacts, two of which can be unlocked at the beginning of the game. The third artifact will require the player to unlock the first excavation tool. This area will have an initial excavation time of 10 seconds and a base chance of success of 35%. The first artifact has a success multiplier of 1x, the second artifact has a success multiplier of 0.8x, and the third artifact has a success multiplier of 0.5x.

The second exhibit will also contain artifacts, only one of which will be attainable upon unlocking the area. The second artifact will require the use of the same tool as the third artifact in area 1, and the third artifact will require the use of a second tool. The initial excavation time will be 20 seconds, and the base success chance will be 15%. The multipliers follow the same structure as the first area. This area will also give much more funds on a failed excavation than the first area.

1.2.2 Objectives

The player's main objective is to recover, restore, and display the six artifacts across the museum's two exhibits. To restore an artifact, the player must collect all its fragments, clean all the fragments, and select the option to assemble the artifact.

The player begins with access to two artifacts in the first exhibit. Completing the artifact generates income and gains the player an increase to reputation, allowing the player to improve the recovery team and unlock the second exhibit and tools to excavate the locked artifact. Exhibit 2 earns the player more funds on a failed excavation but has a much lower chance of successfully excavating a fragment.

The player must earn enough museum funding and reputation to unlock excavation tools. Tool 1 provides access to the third artifact in Exhibit 1 and the second artifact in Exhibit 2. Tool 2 provides access to the final artifact in Exhibit 2.

The final objective is to obtain both excavation tools and assemble all six artifacts. Once every artifact has been placed on display, both exhibits are complete and the player wins the game.

1.2.3 Conflict

The main conflict arises from the fact that the player only has access to one recovery team. The player must choose between a safe, slow approach by finishing the early artifacts to gain enough funds to upgrade the team enough to quickly obtain the harder fragments or try to go to the harder zones first and hope to get lucky with the low fragment success chance and high excavation times.

Area 2 has a lower base fragment-discovery chance and a longer excavation interval than Area 1. However, unsuccessful excavation attempts in Area 2 provide larger amounts of sellable salvage funding. This creates a risk/reward decision: the player may attempt to rush difficult late-game artifacts and earn more funding from failures or remain in the earlier site to complete safer artifacts and purchase upgrades first.

Museum funding is limited and can be spent on recovery-speed upgrades, fragment-discovery upgrades, excavation tools, and archaeological-site access. Spending funding on one option delays progress toward the others. Reputation also creates progression barriers because it can only be earned by completing artifacts.

1.2.4 Menus and Interfaces

1.2.4.1 Shared HUD and navigation

A shared HUD is visible on both the museum exhibit screens and on the excavation map. The HUD displays the player's current museum funds, reputation, and the active artifact excavation's current timer. It also contains a button that opens the pause and settings menu.

The exhibit screen also says the name of the current exhibit as part of the environment.

Navigation Arrows also appear on either side of the screen start, leftmost with the world map, exhibit 1, and exhibit 2. The Left arrow disappears on the world map, and right arrow disappears on exhibit 2 since there are no more screens in those directions. (Figure 7, 8)

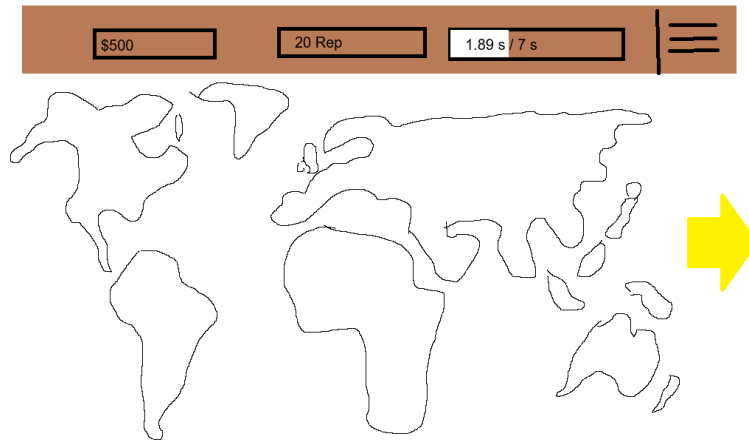


Figure 7. Map w/ Hud

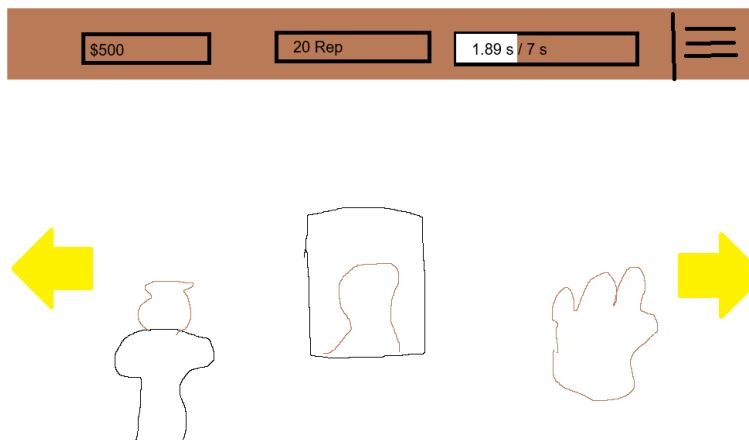


Figure 8. Museum Menu w/ Hud

1.2.4.2 Main Menu

The main menu is shown when the game launches. It displays the name of the game and buttons to start the game, settings, and quit the game. Selecting start begins a new game in exhibit 1, settings opens the settings menu, and quit game exits the game. (Figure 9)

Piece by Piece

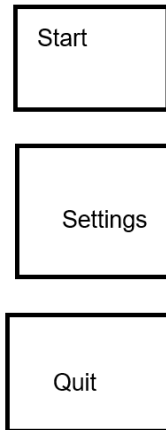


Figure 9. Main Menu

1.2.4.3 Exhibit Screen

There will be 2 exhibit screens. At game start, the second exhibit screen will be a dark background and have a locked icon that says the unlock conditions underneath it. (Figure 10)

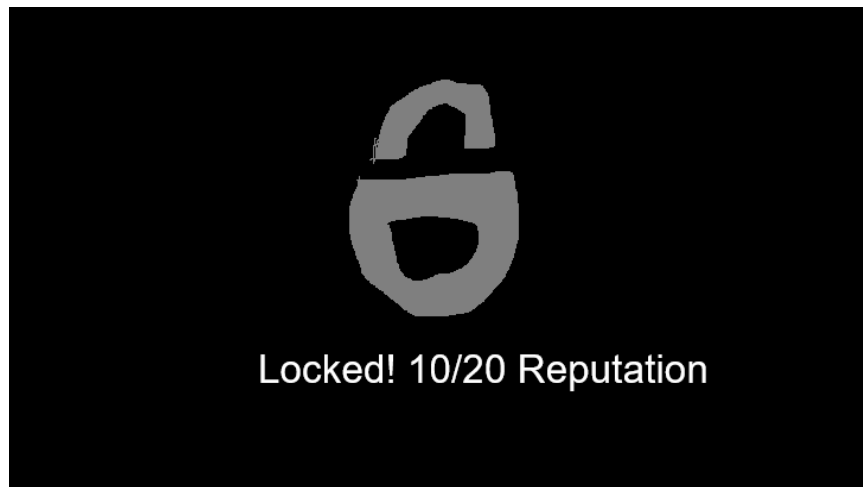


Figure 10. Locked Exhibit

Once exhibit screens are unlocked but not all artifacts are completed, completed artifacts will be displayed on their pedestal or other display method with a bright light above it illuminating the artifact. Unlocked but not finished artifacts will have a case present in the space where the artifact would be displayed without the light. If the player has all available pieces to finish an exhibit, the case will have an exclamation mark above it to let the player

know. Locked artifacts will be dark, and a locked icon will cover the space along with the requirement to unlock the artifact. (Figure 11)



Figure 11. Example Exhibit

1.2.4.4 Excavation Map

The excavation map, accessed by moving left from the first exhibit menu, has a stylized world map with selectable markers for each archaeological site. Available sites use colored map markers, while locked sites are marked in gray. A locked site can still be selected to view its requirements.

Selecting an available site opens the site's information panel, is described below

When a player meets the requirements to unlock a new site, an exclamation mark appears above the marker on the map. The site then enables the user to select the unlock button. (Figure 12)

A site that has all its artifacts completed appears as gold on the map and a gold bordered information panel.

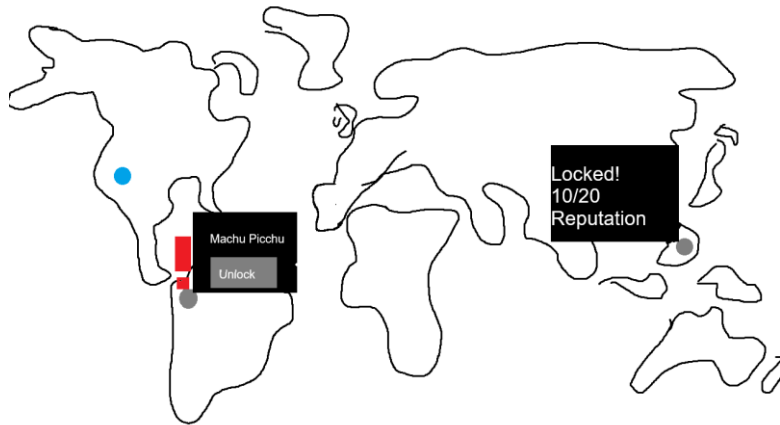


Figure 12. World Map

1.2.4.5 Site information

The site information panel allows the player to select an available artifact as the active target for excavation. The selected artifact has a red border to denote the active one. The panel also displays a progress bar showing the time remaining until the excavation attempt finishes.

The player can change the active artifact at any time by selecting another valid artifact in the panel. Locked artifacts cannot be selected until the player has unlocked the required excavation tool.

The panel also contains two site-specific upgrade buttons for excavation speed and fragment discovery chance. Each button displays the upgrades current effect, next effect, and cost. (Figure 13)

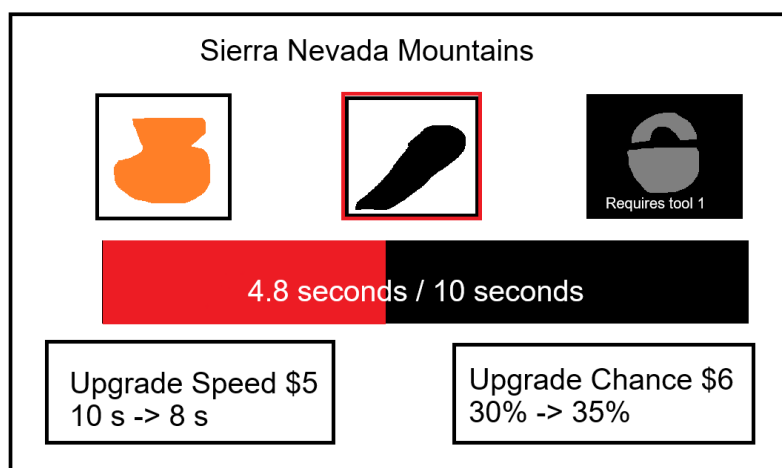


Figure 13. Site Information Panel

1.2.4.6 Excavation Tools

The Excavation map will have a button that opens the equipment panel. This panel shows every unlockable tool in the game, its required reputation amount if still locked and the cost. Tools that are still locked show a locked icon. (Figure 14)

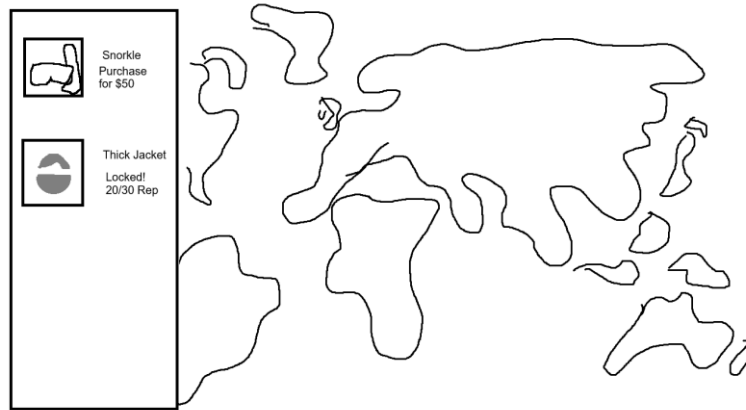


Figure 14. Excavation Tool Panel

1.2.4.7 Pause / Settings

The pause and settings menu is opened from the shared HUD. It has buttons to resume the game, quit the game and open the settings menu. When resumed, the player should return to the previous screen (Figure 15)

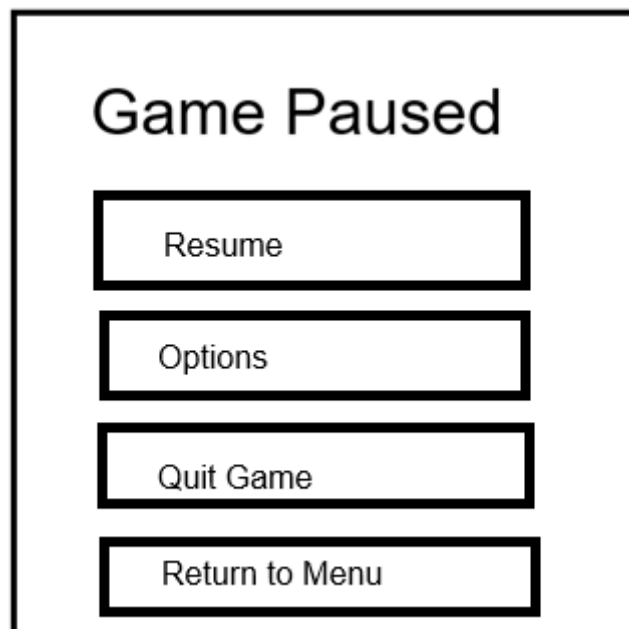


Figure 15. Pause

1.2.4.8 Settings Screen

The settings menu has options for music volume, sound effects volume, a back button, and a short instructional message on how to play the game. The back button should return the player to the previous screen. (Figure 16)

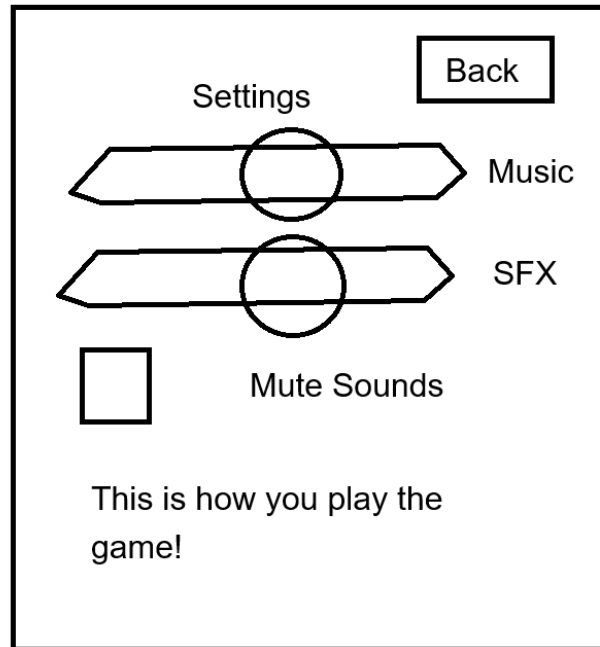


Figure 16. Settings Menu

1.2.4.9 Win Screen

The win screen appears when the player displays all 6 completed artifacts. It congratulates the player and shows the player's final funds, reputation. The buttons allow the player to quit to the main menu, start a new game or exit the game. (Figure 17)

Congratulations! You assembled all 6 artifacts!

Final Funds: \$99999

Final Reputation: 20

Start New Game

Return to Main Menu

Quit Game

Figure 17. Win Screen

2 Technical Design Document and Art List

2.1 Game engine

Piece by Piece will be developed using the Godot Game Engine, which provides 2D rendering, UI controls, scene management tools, input handling, and scripting which is perfect for this kind of game.

The game will use Godot scenes to organize the main menu, exhibits, pause, settings, and excavation map. The mechanics such as the excavation timers, funds, reputation, progress, and button interactions will be implemented with GDScript.

2.2 Other software and Tools

- **Pixelorama:** Free pixel art editor to create artifacts, backgrounds, HUD elements, and other visuals.
 - **Audacity:** Free audio editor used to edit music and sound effects such as button clicks, cleaning sounds, and artifact assembly effects
 - **Git and GitHub:** Git will be used for version control, while the project itself will be stored on GitHub.
- Microsoft Word:** Microsoft Word will be used to write and format the Game Design Document.

2.3 Platform

The game will be developed as a Windows Executable

2.4 Scenes State Diagram

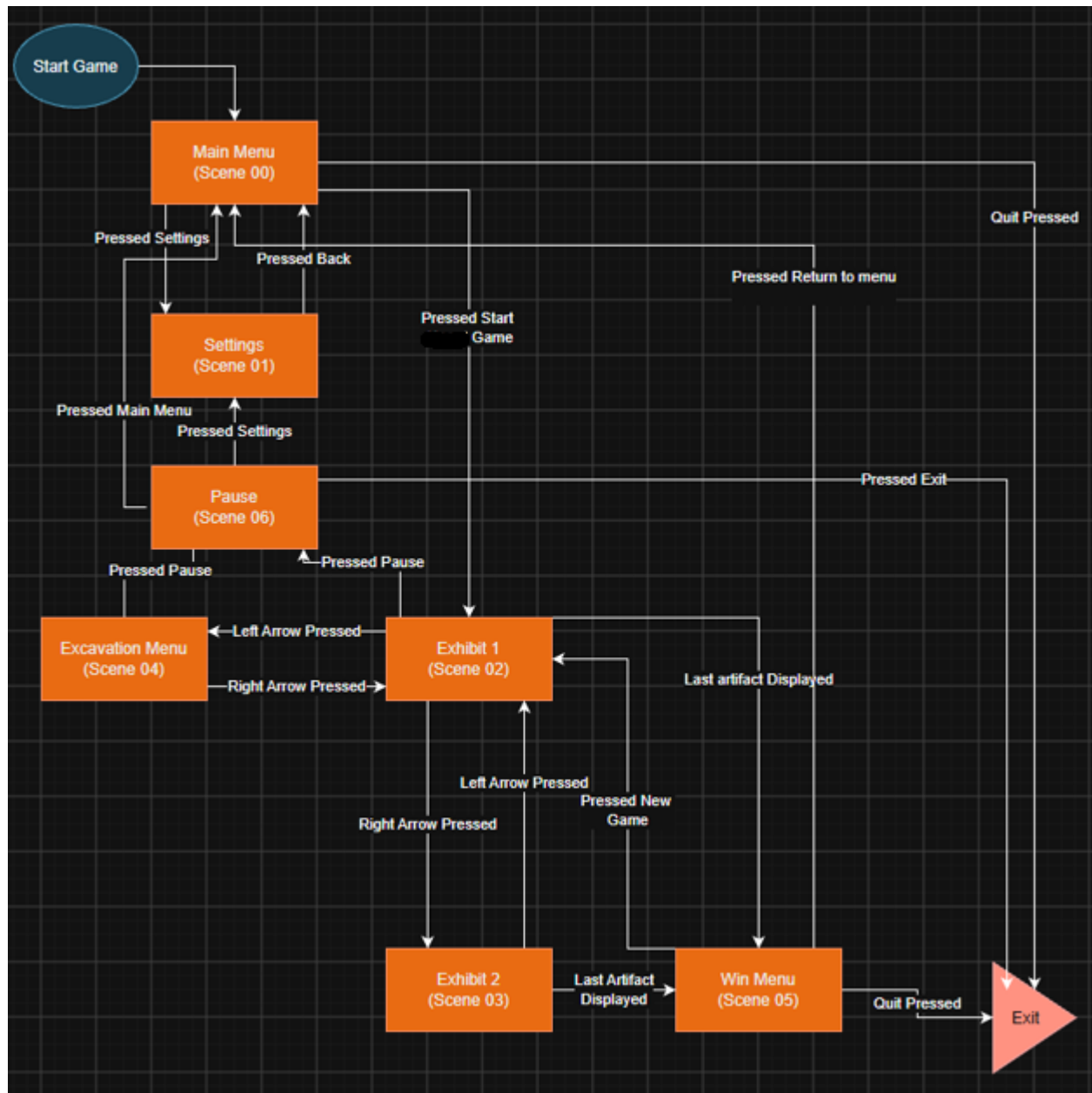


Figure 18. Scene State Diagram

2.5 Scene List

2.5.1 Main Menu

Scene00	
Objectives	Provides the player with a way to start a new game, open settings, or quit the game
Outgoing Transitions	TO Exhibit 1 WHEN player pressed Start TO settings WHEN player presses settings TO exit WHEN player presses Quit Game
Game Objects	Main Menu Manager Main Menu UI Title Text Background Artwork Start Game Button Settings Button Quit Game Button MenuMusic
Comments	n/a

2.5.2 Settings

Scene01	
Objectives	Allows the player to adjust the music and sound-effect volume, mute audio, and view basic tutorials
Outgoing Transitions	TO Main Menu WHEN player presses Back when opening from main menu TO Pause Menu WHEN player presses Back when opening from paused
Game Objects	Settings Manager Settings UI Instructions Label Music Volume Slider SFX Volume Slider Mute Toggle Back Button
Comments	Should return to scene from which player called it

2.5.3 Exhibit 1

Scene02	
Objectives	Displays the first museum exhibit and allows player to view, clean, and assemble the 3 artifacts from Area 1
Outgoing Transitions	TO Excavation Menu WHEN player presses left arrow TO Exhibit 2 WHEN player presses right arrow TO Pause Menu WHEN player presses pause button TO Win Menu WHEN final artifact displayed
Game Objects	Exhibit 1 Manager Exhibit Background CanvasLayer HUD Right Arrow Left Arrow Pause Button Display Areas Fragment Cases Assemble Artifact Button Locked Artifact Display Overlay Artifact Sprites
Comments	N/A

2.5.4 Exhibit 2

Scene03	
Objectives	Displays the second museum exhibit. Before the exhibit is unlocked, the scene shows an overlay showing the exhibit is locked with its requirements. After unlocking, the player can view, clean and assemble the 3 artifacts from Area 2
Outgoing Transitions	TO Exhibit 1 WHEN player presses left arrow TO Pause Menu WHEN player presses pause button TO Win Menu WHEN final artifact displayed
Game Objects	Exhibit 2 Manager Exhibit Background CanvasLayer HUD Left Arrow Pause Button Display Areas Fragment Cases Assemble Artifact Button Locked Artifact Display Overlay Artifact Sprites Locked Exhibit Overlay Unlock Requirement Label
Comments	N/A

2.5.5 Excavation Menu

Scene04	
Objectives	Allows the player to select an archaeology site and active artifact, purchase site upgrades, and open the excavation tools panel.
Outgoing Transitions	TO Exhibit 1 WHEN player presses right arrow TO Pause Menu WHEN player presses pause button
Game Objects	Excavation Menu Manager World Map Background Site Markers CanvasLayer HUD Site Information Panel Active Artifact Buttons Excavation Progress Bar Speed Upgrade Button Discovery Chance Upgrade Button Excavation Tools Panel Right Arrow Pause Button
Comments	Site Information Panel and Excavation Tools panel within this scene

2.5.6 Win Menu

Scene05	
Objectives	Congratulates the player for restoring all artifacts and displays final museum funds, and reputation
Outgoing Transitions	TO Exhibit 1 WHEN player presses New Game TO Main Menu WHEN player presses Main Menu TO Exit WHEN player presses Quit Game
Game Objects	Win Menu Manager Win Menu UI Completed Exhibit Background Completion Label Final Funding Label Final Reputation Label New Game Button Main Menu Button Quit Game Button
Comments	Should only be reached after completing all six artifacts

2.5.7 Pause Menu

Scene06	
Objectives	Pauses gameplay and provides options to resume, open settings, quit, and return to main menu
Outgoing Transitions	TO previous scene WHEN player presses Resume TO Settings WHEN player presses Settings TO Main Menu WHEN player presses Return to Main Menu TO Exit WHEN player presses Quit
Game Objects	Pause Menu Manager Pause Menu UI Darkened Background Overlay Resume Button Settings Button Quit Button Return to Main Menu Button
Comments	Can be opened from Exhibit 1, Exhibit 2, or the Excavation Menu

2.6 Game Object / Component Diagrams

2.6.1 Main Menu

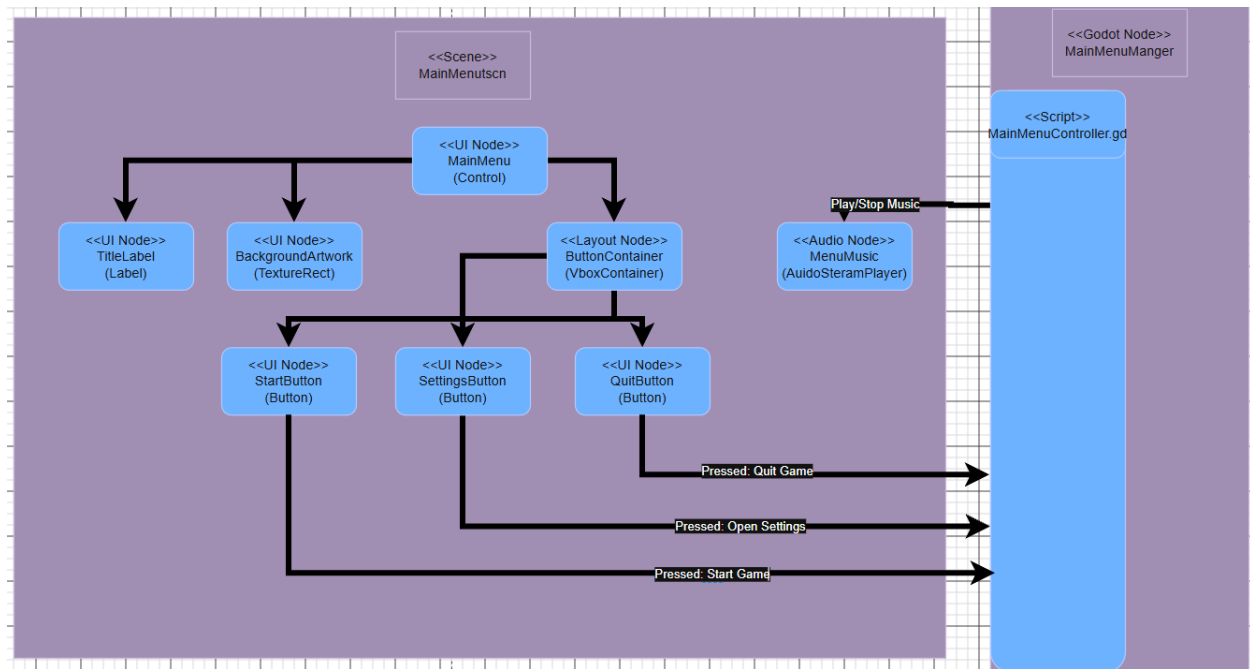


Figure 19. Main Menu Game Object / Component Diagram

2.6.2 Exhibit 1

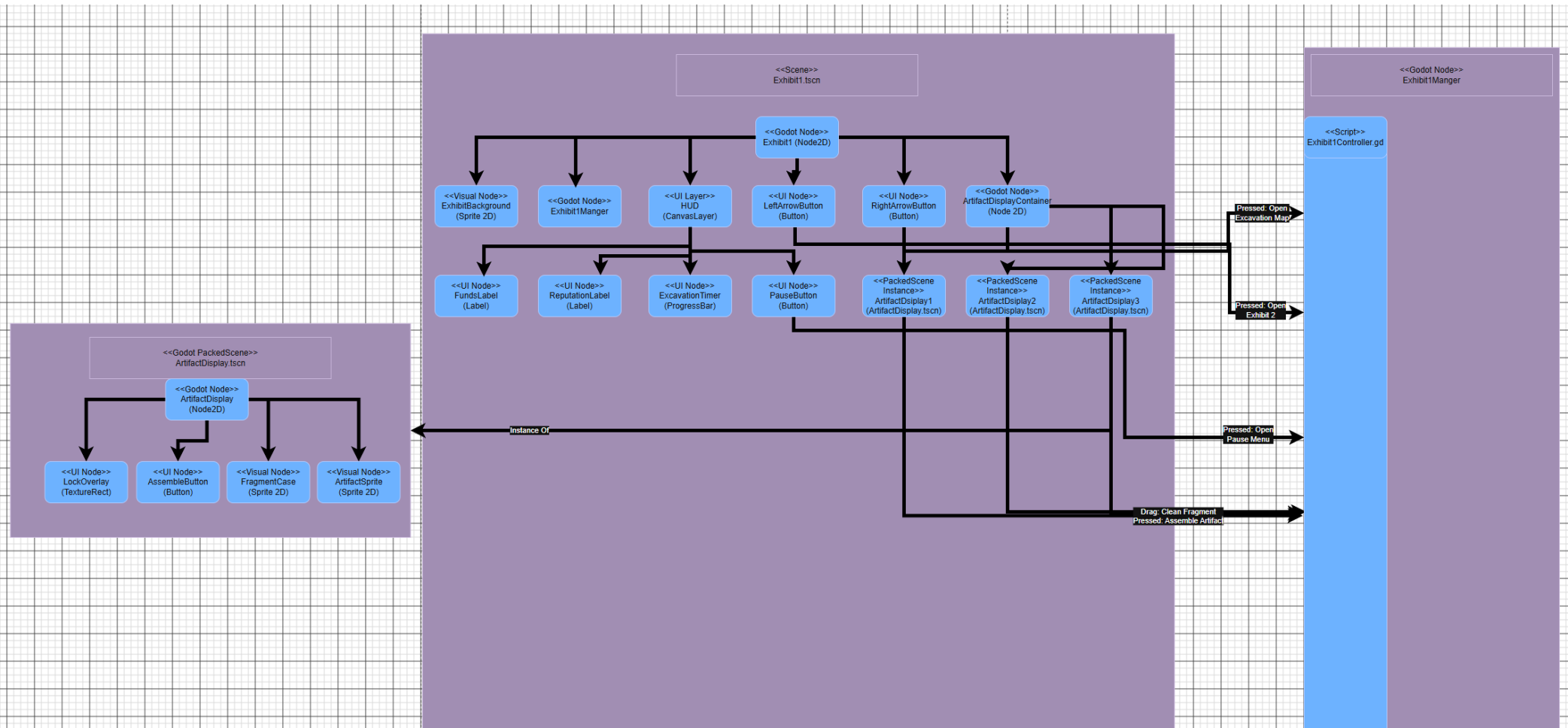


Figure 20. Exhibit 1 Game Object / Component Diagram

2.6.3 Exhibit 2

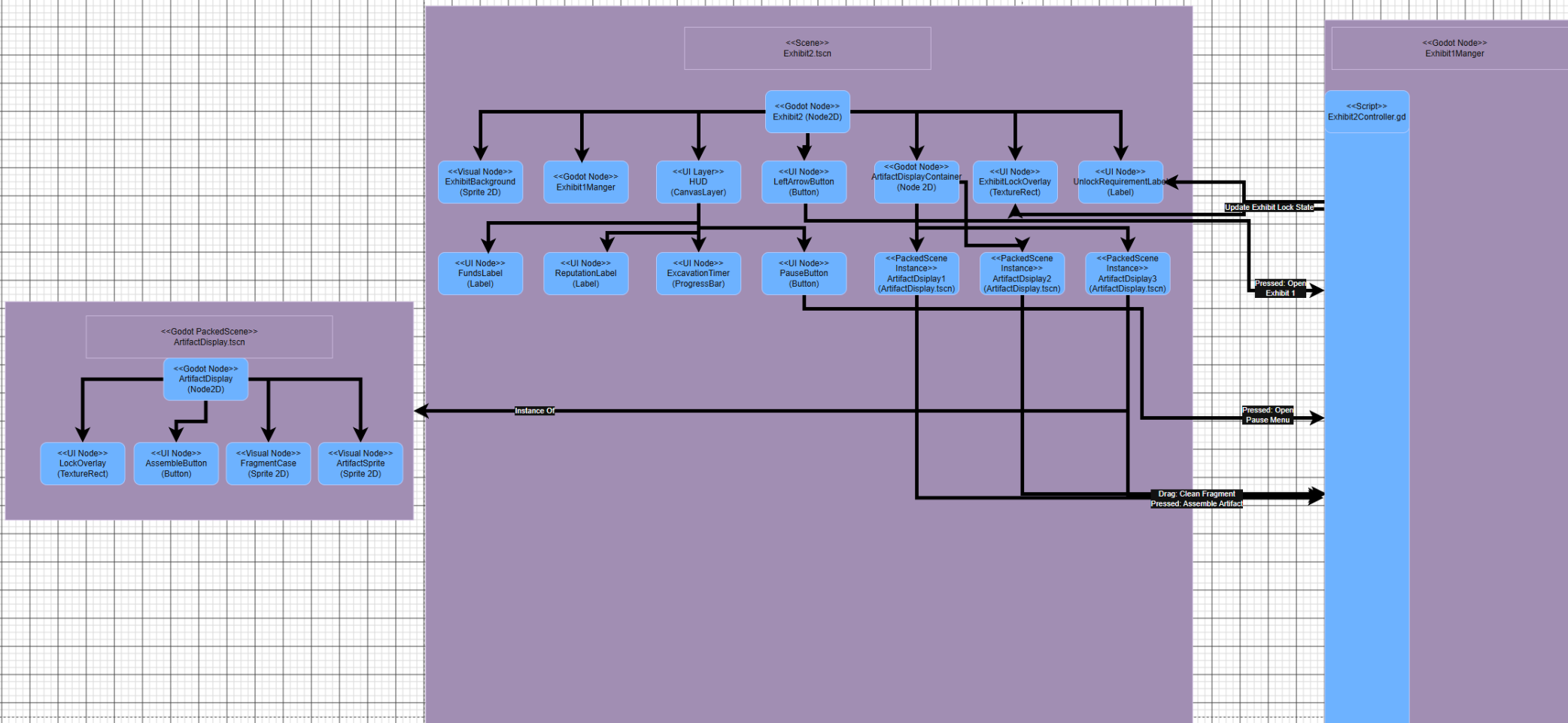


Figure 21. Exhibit 2 Game Object / Component Diagram

2.6.4 Excavation Menu

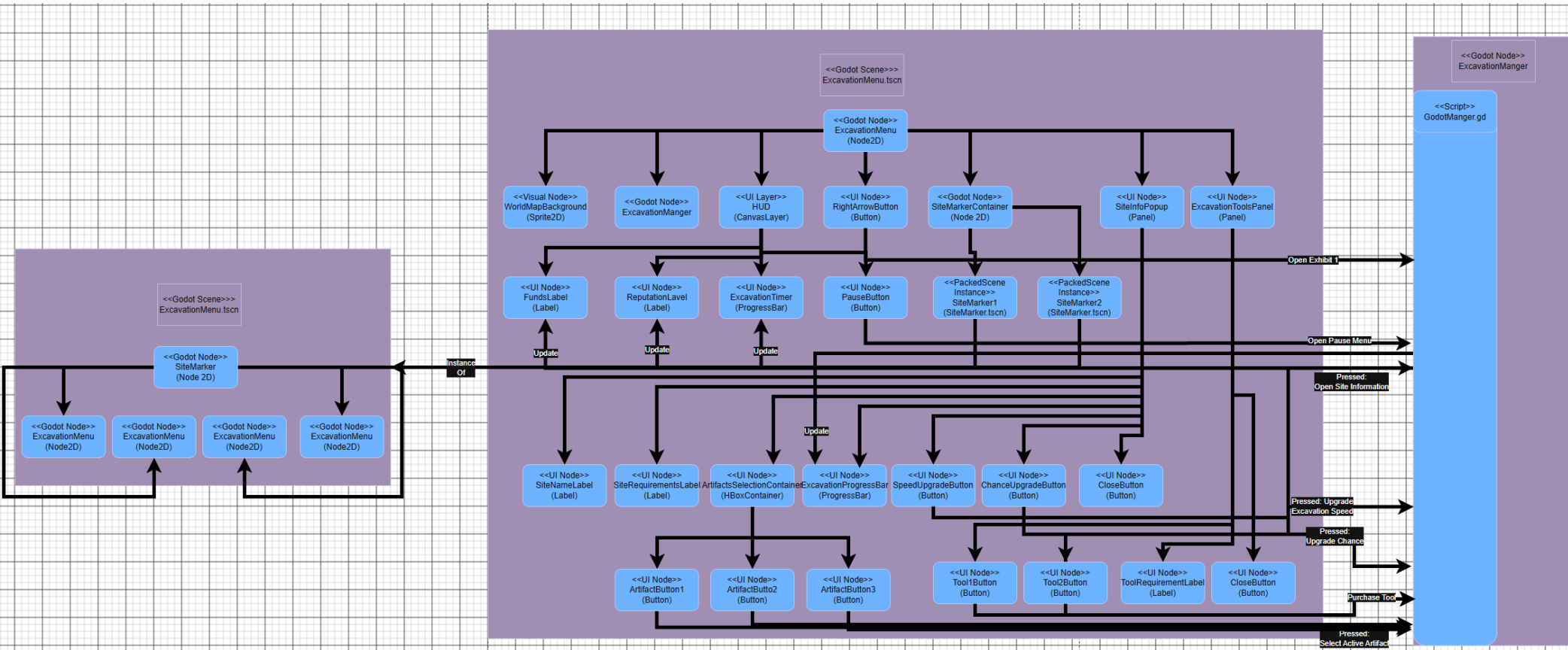


Figure 22. Excavation Menu Game Object / Component Diagram

2.6.5 Win Menu

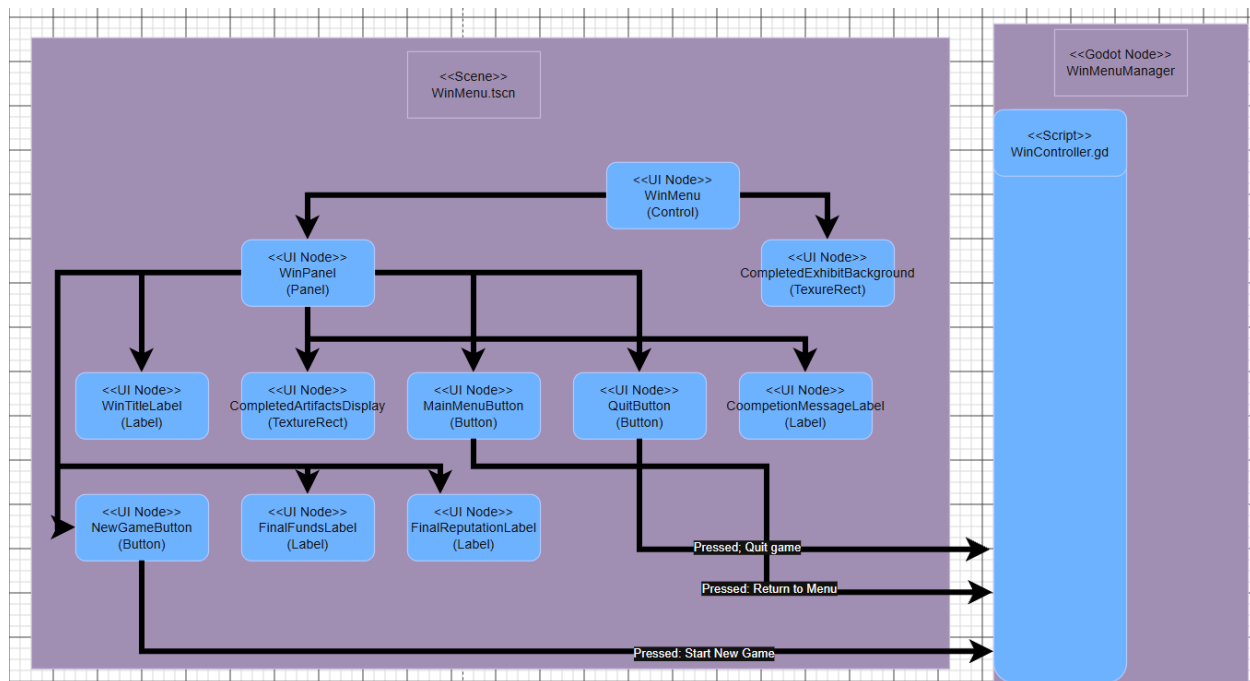


Figure 23. Win Menu Game Object / Component Diagram

2.6.6 Pause Menu

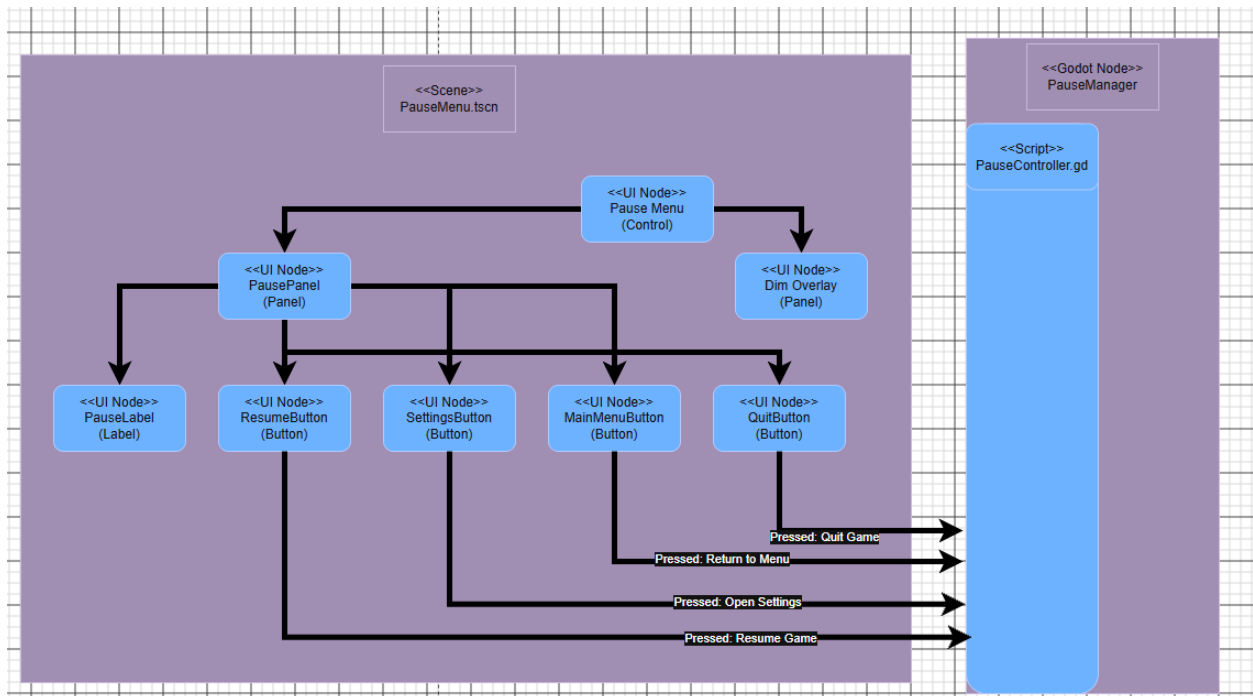


Figure 24. Pause Menu Game Object / Component Diagram

2.6.7 Settings Menu

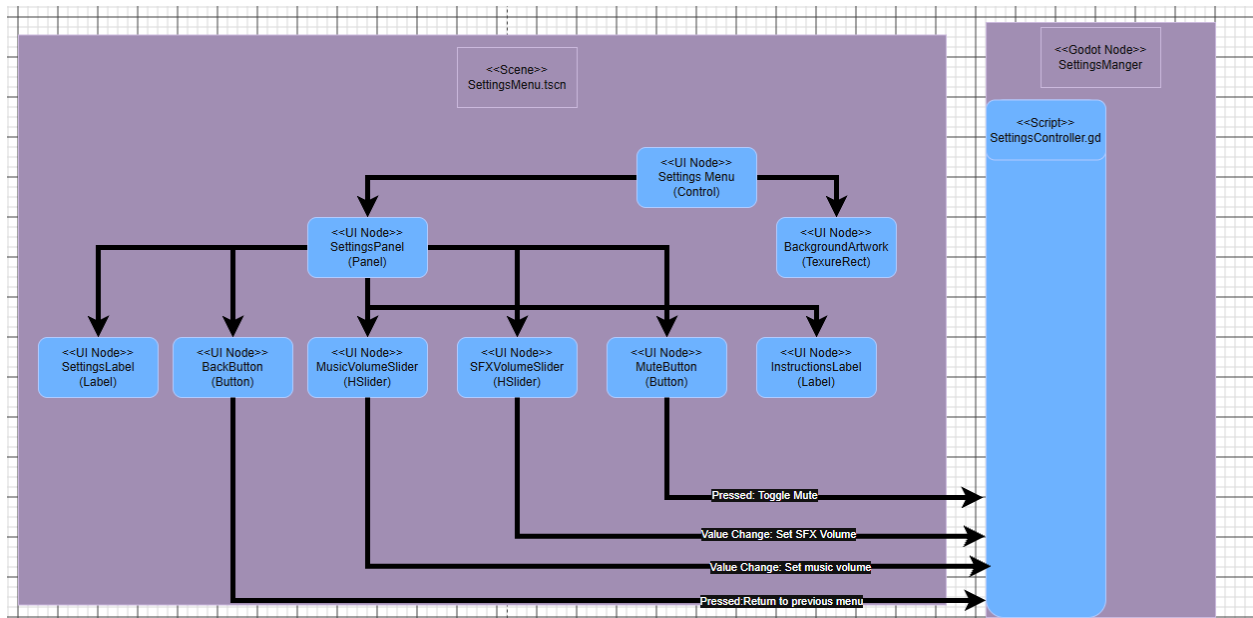


Figure 25. Settings Menu Game Object / Component Diagram

2.7 Game Object List

2.7.1 Game State

Game State (Autoload)	
Description	Stores the player's funds, reputation, artifact progress, unlocked tools, site unlocks, and active excavation data across scenes.
Tags	Game Data; Persistent
Required Components	Node GameState.gd
Related Game Objects	
Parent	n/a - Autoload
Children	n/a
Associations	All scene managers
Comments	Keeps progress when the player changes scenes.

2.7.2 Main Menu Manager

Main Menu Manager	
Description	Controls the Main Menu scene and processes menu-button input.
Tags	Menu; Manager
Required Components	Node MainMenuController.gd
Related Game Objects	
Parent	MainMenu

Children	n/a
Associations	Main Menu UI Settings Menu Exhibit 1 Game State
Comments	Opens the selected menu, starts a new game, or exits the game.

2.7.3 Main Menu UI

Main Menu UI	
Description	The visual Main Menu interface.
Tags	UI; Menu
Required Components	Control TextureRect Label VBoxContainer
Related Game Objects	
Parent	MainMenu
Children	TitleLabel BackgroundArtwork ButtonContainer StartButton SettingsButton QuitButton
Associations	MainMenuManager
Comments	ButtonContainer organizes the menu buttons vertically.

2.7.4 Menu Music

Menu Music	
Description	Plays the background music used on the Main Menu.
Tags	Audio; Menu
Required Components	AudioStreamPlayer Menu music audio file
Related Game Objects	
Parent	MainMenu
Children	n/a
Associations	MainMenuManager Settings Menu audio controls
Comments	Volume can be changed through the Settings Menu.

2.7.5 Settings Manager

Settings Manager	
Description	Controls settings changes and returns the player to the menu that opened Settings.
Tags	Menu; Manager; Settings
Required Components	Node SettingsController.gd
Related Game Objects	
Parent	SettingsMenu

Children	n/a
Associations	Settings UI Main Menu Pause Menu Audio buses
Comments	Applies music and sound-effect volume changes.

2.7.6 Settings UI

Settings UI	
Description	Displays the settings controls and basic instructions.
Tags	UI; Settings
Required Components	Control Panel Label HSlider Button
Related Game Objects	
Parent	SettingsMenu
Children	BackgroundArtwork SettingsPanel SettingsLabel MusicVolumeSlider SFXVolumeSlider MuteButton InstructionsLabel BackButton
Associations	SettingsManager

Comments	BackButton returns to either Main Menu or Pause Menu.
-----------------	---

2.7.7 Exhibit 1 Manager

Exhibit 1 Manager	
Description	Controls Exhibit 1 artifact cleaning, assembly, passive donations, and navigation.
Tags	Gameplay; Manager; Exhibit 1
Required Components	Node Exhibit1Controller.gd
Related Game Objects	
Parent	Exhibit1
Children	n/a
Associations	HUD ArtifactDisplay Game State Exhibit 2 Excavation Menu Pause Menu
Comments	Controls the first three artifacts and the Area 1 exhibit.

2.7.8 Exhibit 1 Background

Exhibit 1 Background	
Description	The pixel-art museum environment displayed behind Exhibit 1's artifacts.
Tags	Visual; Museum; Exhibit 1

Required Components	Sprite2D Exhibit background texture
Related Game Objects	
Parent	Exhibit1
Children	n/a
Associations	Exhibit1Manager
Comments	Contains the visual exhibit cases and museum setting.

2.7.9 HUD

HUD	
Description	Displays the player's funds, reputation, active excavation timer, and pause control.
Tags	UI; Gameplay
Required Components	CanvasLayer Control Label ProgressBar Button
Related Game Objects	
Parent	Exhibit 1, Exhibit 2, or Excavation Menu
Children	FundsLabel ReputationLabel ExcavationTimer PauseButton

Associations	Scene managers Game State Pause Menu
Comments	Reused in all gameplay scenes.

2.7.10 Exhibit Navigation Buttons

Exhibit Navigation Buttons	
Description	Allows the player to move between museum exhibits and the Excavation Menu.
Tags	UI; Navigation
Required Components	Button Arrow-button texture
Related Game Objects	
Parent	Exhibit 1, Exhibit 2, or Excavation Menu
Children	LeftArrowButton and/or RightArrowButton
Associations	Exhibit1Manager Exhibit2Manager ExcavationManager
Comments	Only valid navigation arrows are visible in each scene.

2.7.11 Artifact Display

Artifact Display (Godot PackedScene)	
Description	A reusable display case for a single artifact and its fragments.
Tags	Gameplay; Artifact; Museum

Required Components	Node2D Artifact display controller connection
Related Game Objects	
Parent	ArtifactDisplayContainer
Children	ArtifactSprite FragmentCase AssembleButton LockOverlay
Associations	Exhibit1Manager Exhibit2Manager Game State
Comments	ArtifactDisplay.tscn. Instanced six times, once for each artifact.

2.7.12 Artifact Sprite

Artifact Sprite	
Description	Shows an artifact's visual state, from fragments to a completed museum display.
Tags	Visual; Artifact
Required Components	Sprite2D Artifact sprite-sheet or texture
Related Game Objects	
Parent	ArtifactDisplay
Children	n/a

Associations	ArtifactDisplay Exhibit manager
Comments	Uses the completed-artifact sprite after the assembly animation.

2.7.13 Fragment Case

Fragment Case	
Description	Holds uncleaned and cleaned artifact fragments before assembly.
Tags	Visual; Gameplay; Artifact
Required Components	Sprite2D Cleaning interaction area
Related Game Objects	
Parent	ArtifactDisplay
Children	n/a
Associations	ArtifactDisplay Exhibit manager
Comments	Fragments earn small passive donations before assembly.

2.7.14 Assemble Button

Assemble Button	
Description	Starts the artifact assembly animation after all fragments are cleaned.
Tags	UI; Gameplay; Artifact

Required Components	Button
Related Game Objects	
Parent	ArtifactDisplay
Children	n/a
Associations	ArtifactDisplay Exhibit manager Game State
Comments	Disabled until every collected fragment is clean.

2.7.15 Exhibit 2 Manager

Exhibit 2 Manager	
Description	Controls Exhibit 2's lock state, artifact displays, cleaning, assembly, and navigation.
Tags	Gameplay; Manager; Exhibit 2
Required Components	Node Exhibit2Controller.gd
Related Game Objects	
Parent	Exhibit2
Children	n/a
Associations	HUD ArtifactDisplay Game State

	Exhibit 1 Pause Menu
Comments	Controls the three Area 2 artifacts.

2.7.16 Exhibit 2 Lock UI

Exhibit 2 Lock UI	
Description	Shows that Exhibit 2 is locked and displays the requirement needed to unlock it.
Tags	UI; Lock; Exhibit 2
Required Components	TextureRect Label
Related Game Objects	
Parent	Exhibit2
Children	ExhibitLockOverlay UnlockRequirementLabel
Associations	Exhibit2Manager Game State
Comments	The player may view this scene before it is unlocked but cannot interact with artifacts.

2.7.17 Excavation Manager

Excavation Manager	
Description	Controls the map, selected site, active artifact, excavation upgrades, and Excavation Tool purchases.

Tags	Gameplay; Manager; Excavation
Required Components	Node ExcavationController.gd
Related Game Objects	
Parent	ExcavationMenu
Children	n/a
Associations	World Map Site Marker Site Information Panel Excavation Tools Panel HUD Game State
Comments	Updates site locks, alerts, timers, and artifact availability.

2.7.18 World Map

World Map	
Description	Displays the stylized map containing the two archaeological-site markers.
Tags	Visual; Gameplay; Map
Required Components	Sprite2D World-map texture
Related Game Objects	
Parent	ExcavationMenu
Children	SiteMarkerContainer

Associations	ExcavationManager Site Marker
Comments	Acts as the Excavation Menu background.

2.7.19 Site Marker

Site Marker (Godot PackedScene)	
Description	A reusable clickable world-map marker representing one archaeological site.
Tags	Gameplay; Map; Excavation
Required Components	Node2D TextureButton Sprite2D
Related Game Objects	
Parent	SiteMarkerContainer
Children	MarkerButton MarkerSprite LockOverlay AlertIcon
Associations	ExcavationManager Site Information Panel Game State
Comments	SiteMarker.tscn. Instanced twice, once for each site.

2.7.20 Site Information Panel

Site Information Panel

Description	Shows the selected site's requirements, three artifact choices, excavation progress, and upgrade controls.
Tags	UI; Gameplay; Excavation
Required Components	Panel Label HBoxContainer TextureButton ProgressBar Button
Related Game Objects	
Parent	ExcavationMenu
Children	SiteNameLabel SiteRequirementsLabel ArtifactSelectionContainer ArtifactButton1-3 ExcavationProgressBar SpeedUpgradeButton ChanceUpgradeButton CloseButton
Associations	ExcavationManager Site Marker Game State
Comments	The player chooses the active artifact and purchases that site's upgrades here.

2.7.21 Excavation Tools Panel

Excavation Tools Panel	
Description	Lets the player purchase Excavation Tool 1 and Excavation Tool 2.

Tags	UI; Gameplay; Tools
Required Components	Panel Button Label
Related Game Objects	
Parent	ExcavationMenu
Children	Tool1Button Tool2Button ToolRequirementLabel CloseButton
Associations	ExcavationManager Game State Exhibit 1 Exhibit 2
Comments	Tools require sufficient funds and reputation.

2.7.22 Pause Menu Manager

Pause Menu Manager	
Description	Pauses gameplay and controls the actions selected on the Pause Menu.
Tags	Menu; Manager; Pause
Required Components	Node PauseController.gd
Related Game Objects	
Parent	PauseMenu

Children	n/a
Associations	Pause Menu UI Settings Menu Main Menu Current gameplay scene Game State
Comments	Can be opened from Exhibit 1, Exhibit 2, or the Excavation Menu.

2.7.23 Pause Menu UI

Pause Menu UI	
Description	Provides controls to resume, open settings, return to the Main Menu, or quit while gameplay is paused.
Tags	UI; Menu; Pause
Required Components	Control Panel ColorRect or Panel Label Button
Related Game Objects	
Parent	PauseMenu
Children	DimOverlay PausePanel PauseLabel ResumeButton SettingsButton MainMenuButton QuitButton

Associations	PauseMenuManager
Comments	DimOverlay darkens the gameplay scene behind the menu.

2.7.24 Win Menu Manager

Win Menu Manager	
Description	Controls the completion screen after all six artifacts have been assembled.
Tags	Menu; Manager; Win
Required Components	Node WinController.gd
Related Game Objects	
Parent	WinScreen
Children	n/a
Associations	Win Menu UI Game State Exhibit 1 Main Menu
Comments	Opens a new game, returns to Main Menu, or quits.

2.7.25 Win Menu UI

Win Menu UI	
Description	Displays the completed museum, final player progress, and post-game controls.
Tags	UI; Menu; Win

Required Components	Control Panel TextureRect Label Button
Related Game Objects	
Parent	WinScreen
Children	BackgroundArtwork WinPanel WinTitleLabel CompletedArtifactsDisplay CompletionMessageLabel FinalFundsLabel FinalReputationLabel NewGameButton MainMenuButton QuitButton
Associations	WinMenuManager Game State
Comments	Only appears after all six artifacts are assembled.

2.8 Component List

2.8.1 GameState.gd

GameState.gd (Custom)	
Description	Stores persistent game data, including funds, reputation, artifact progress, tools, site unlocks, and the active excavation.
Requires	Node

Comments	Configured as a Godot Autoload so every scene manager can access the same progress data.
-----------------	--

2.8.2 MainMenuController.gd

MainMenuController.gd (Custom)	
Description	Handles Main Menu button input and changes to the selected scene.
Requires	MainMenu Game State
Comments	Starts a new game, opens Settings, and quits the application.

2.8.3 SettingsController.gd

SettingsController.gd (Custom)	
Description	Controls the settings menu and applies the player's audio preferences.
Requires	SettingsMenu Godot audio buses
Comments	Updates music volume, sound-effect volume, mute state, and Back button destination.

2.8.4 Exhibit1Controller.gd

Exhibit1Controller.gd (Custom)	
Description	Controls Exhibit 1 fragment cleaning, artifact assembly, passive donations, and navigation.
Requires	Exhibit1 ArtifactDisplay instances Game State

Comments	Updates the HUD and locks the third Area 1 artifact until Tool 1 is purchased.
-----------------	--

2.8.5 Exhibit2Controller.gd

Exhibit2Controller.gd (Custom)	
Description	Controls Exhibit 2 lock state, artifact displays, cleaning, assembly, and navigation.
Requires	Exhibit2 ArtifactDisplay instances Game State
Comments	Enables the exhibit after its unlock requirement is met and applies tool restrictions.

2.8.6 ExcavationController.gd

ExcavationController.gd (Custom)	
Description	Controls site selection, active-artifact selection, excavation progress, upgrades, and Excavation Tool purchases.
Requires	ExcavationMenu SiteMarker instances SiteInfoPopup ExcavationToolsPanel Game State
Comments	Updates marker locks, alert icons, artifact availability, funds, reputation, and timers.

2.8.7 PauseController.gd

PauseController.gd (Custom)

Description	Pauses and resumes gameplay and handles Pause Menu controls.
Requires	PauseMenu Current gameplay scene Game State
Comments	Opens Settings, returns to Main Menu, resumes gameplay, or quits the game.

2.8.8 WinController.gd

WinController.gd (Custom)	
Description	Controls the Win Screen after all six artifacts are assembled.
Requires	WinScreen Game State
Comments	Displays final progress and handles New Game, Main Menu, and Quit buttons.

2.8.9 Node

Node (Provided by Godot)	
Description	A basic Godot node used for non-visual organization and manager objects.
Requires	Godot Engine
Comments	Used for GameState and scene managers that do not need a visual position or interface.

2.8.10 Node2D

Node2D (Provided by Godot)

Description	A 2D node with position, rotation, and scale.
Requires	Godot Engine
Comments	Used as the root for exhibits, Excavation Menu, ArtifactDisplay, and SiteMarker objects.

2.8.11 Control

Control (Provided by Godot)	
Description	A base node for interface elements that supports anchors, layout, and input.
Requires	Godot Engine
Comments	Used as the root for menu interfaces and as the basis for UI elements.

2.8.12 CanvasLayer

CanvasLayer (Provided by Godot)	
Description	Draws child interface elements above the 2D game world.
Requires	Godot Engine
Comments	Used for the HUD so its values remain visible over gameplay scenes.

2.8.13 Sprite2D

Sprite2D (Provided by Godot)	
Description	Displays a 2D texture or sprite in the game world.
Requires	Godot Engine Image texture or sprite sheet

Comments	Used for museum backgrounds, artifact sprites, fragment cases, map markers, and alert icons.
-----------------	--

2.8.14 TextureRect

TextureRect (Provided by Godot)	
Description	Displays and scales a texture as a user-interface element.
Requires	Godot Engine Image texture
Comments	Used for menu artwork and locked-overlay visuals.

2.8.15 Label

Label (Provided by Godot)	
Description	Displays text in the user interface.
Requires	Godot Engine Font resource
Comments	Used for titles, funds, reputation, requirements, instructions, and completion messages.

2.8.16 Button and TextureButton

Button and TextureButton (Provided by Godot)	
Description	Interface controls that emit a pressed signal when clicked. Button uses text; TextureButton uses an image.
Requires	Godot Engine
Comments	Used for menus, navigation arrows, artifact selections, upgrades, tools, pause controls, and assembly actions.

2.8.17 Panel and ColorRect

Panel and ColorRect (Provided by Godot)	
Description	Interface elements used to create popup windows, menu backgrounds, and screen-darkening overlays.
Requires	Godot Engine
Comments	Panels are used for site information, tools, settings, pause, and win menus.

2.8.18 VBoxContainer and HBoxContainer

VBoxContainer and HBoxContainer (Provided by Godot)	
Description	Layout nodes that automatically arrange child controls vertically or horizontally.
Requires	Godot Engine
Comments	VBoxContainer organizes Main Menu buttons; HBoxContainer organizes artifact-selection buttons.

2.8.19 ProgressBar

ProgressBar (Provided by Godot)	
Description	Displays a value as a visual progress bar.
Requires	Godot Engine
Comments	Used for the active excavation timer and selected-site progress.

2.8.20 HSlider

HSlider (Provided by Godot)

Description	Lets the player choose a numeric value by dragging a slider handle.
Requires	Godot Engine
Comments	Used for music and sound-effect volume settings.

2.8.21 AudioStreamPlayer

AudioStreamPlayer (Provided by Godot)	
Description	Plays an audio file in a Godot scene.
Requires	Godot Engine Audio stream file
Comments	Used to play Main Menu music and controlled through Settings.

2.8.22 PackedScene

PackedScene (Provided by Godot)	
Description	A reusable saved Godot scene that can be instanced multiple times.
Requires	Godot Engine .tscn scene file
Comments	Used for ArtifactDisplay.tscn and SiteMarker.tscn.

2.8.23 AnimationPlayer

AnimationPlayer (Provided by Godot)	
Description	Plays timed property and sprite animations.
Requires	Godot Engine Animation tracks

Comments	Used for the artifact assembly animation, where fragments rise, rotate, snap together, and drop into the display.
-----------------	---

2.9 Art List

2.9.1 Static 2D Assets

2.9.1.1 Main Menu Background

Main Menu Background	
Description	Pixel art title image for Piece by Piece. It shows the museum building against a blue sky and is used on the title page and Main Menu.
Format	PNG RGBA, 1920 x 1080p
Comments	Static Background Menu Buttons and title render above this image
Concept Art	Figure 10, Overview image

2.9.1.2 Main Menu Title Splash

Main Menu Title Splash	
Description	Stylized font for main menu displaying the name of the game.
Format	PNG RGBA, approx 512 x 256p
Comments	Static image with transparent background Rendered over the main menu background
Concept Art	Figure 10, Overview Image

2.9.1.3 Exhibit Backgrounds

Exhibit Backgrounds	
Description	Pixel art Museum rooms containing pedestals, artifact cases, lighting, and exhibit overlays. There should be one for each of the 2 rooms in different styles.
Format	PNG RGBA 1920 x 1080p. Two files
Comments	Static background art
Concept Art	Figure 8

2.9.1.4 World Map Background

World Map Background	
Description	Stylized pixel art world map for the excavation menu. Displays locations of the archaeological sites.
Format	PNG RGBA, 1920 x 1080p
Comments	Static local background Site markers and popup menus render over map
Concept Art	Figure 7

2.9.1.5 Fragment Case Sprites

Fragment Case Sprites	
Description	Pixel art display cases that hold the uncleaned and cleaned artifact fragments before artifact assembly
Format	PNG RGBA, 128 x 128p
Comments	Static Local Sprite. A case may show fragments, empty spots, or an exclamation mark above the case when ready to assemble. Should be a variation for each of the 6 artifacts. Can use the same sprite when closed on the ground
Concept Art	Figure 4

2.9.1.6 Pedestal Sprites

Pedestal Sprites	
Description	Pixel art museum pedestals used to display fully assembled artifacts
Format	PNG RGBA, 128 x 128p
Comments	Static Local sprite. Completed artifacts per 2.9.2.1 should be placed on top. Should be unique and match each of the 6 artifacts.
Concept Art	

2.9.1.7 Light Overlay Sprite

Light Overlay Sprite	
Description	A transparent pixel art glow overlay placed above a completed artifact to show a completed display
Format	PNG RGBA with transparent background 128 x 128p
Comments	Hidden for incomplete artifacts
Concept Art	

2.9.1.8 Site Marker Sprite Sheet

Site Marker Set	
Description	Clickable pixel art markers representing archaeological sites on the world map. 3 variants for locked, available and completed states.
Format	PNG RGBA sprite sheet, 32 x 32p cells
Comments	Local asset used by the reusable SiteMarker PackedScene. Includes lock, alert, and completed-gold variants
Concept Art	Figure 12

2.9.1.9 Excavation Tool Icons

Excavation Tool Icons	
Description	Pixel art icons for Excavation tool 1 and Excavation tool 2, including the locked and purchased states
Format	PNG RGBA, 32 x 32p
Comments	Static local icons shown in the Excavation Tools Panel and artifact requirement displays.
Concept Art	Figure 14

2.9.1.10 HUD and UI Icon Set

HUD and UI Icon Set	
Description	Pixel-art icons and UI decorations for funds, reputation, excavation timer, pause, navigation arrows, locks, alerts, and artifact-selection borders.
Format	PNG RGBA; 16 x 16, 24 x 24, and 32 x 32 pixel icons
Comments	Static local interface assets. Selection and completion borders may also be created through Godot UI styling.
Concept Art	Figures 7, 8, 10, and 14

2.9.1.11 UI Panels and Buttons

UI Panels and Buttons	
Description	Pixel-art panels and buttons used for menus, settings, pause controls, site information, tool purchases, and the Win Screen.
Format	PNG RGBA nine-slice panels; 32 x 32 to 128 x 128 base textures
Comments	Local UI assets. Buttons include normal, hovered, disabled, locked, and selected states where needed.
Concept Art	Figures 10, 14, 15, and 16

2.9.1.12 Win Screen Background / Completion Art

Win Screen Background / Completion Art	
Description	Pixel-art final museum image showing a completed exhibit and restored artifacts behind the Win Screen controls.
Format	PNG RGBA; 1920 x 1080 pixels
Comments	Static local background used with the completion message, final funds, final reputation, and Win Screen buttons.
Concept Art	Figure 16. Win Screen

2.9.2 Animated 2D Assets

2.9.2.1 Artifact Assembly Animation

Artifact Assembly Animation	
Description	A local pixel-art animation that plays when the player presses Assemble after cleaning every fragment of an artifact. The pieces rise from the case, rotate, snap together, and drop onto the pedestal as a completed artifact.
Format	PNG RGBA sprite sheet; 64 x 64 pixel cells; played through Godot AnimationPlayer
Animated Sequence	1. Uncleaned Idle: fragments sit in the case with a subtle dust or shimmer effect. 2. Cleaning: fragments brighten as the player cleans them. 3. Assembly: fragments rise, rotate, snap together, and drop onto the pedestal.

	4. Completed Display: the restored artifact has a brief shine/light animation.
Comments	
Concept Art	

2.10 Audio Assets

2.10.1 Music Assets

2.10.1.1 Museum Theme

Museum Theme	
Description	A calm looping instrumental tract that plays in the main menu, museum exhibits, and excavation.
Format	OGG, Stereo, 1-2 minutes, seamless loop
Comments	Probably use midi sounds to complement the pixel art
Concept Art	

2.10.2 Sound Effect Assets

2.10.2.1 UI Button Click

UI Button Click	
Description	A short click sounds when the player selects a button, arrow, marker, or menu option
Format	OGG, mono, under a second
Comments	Used through menus Maybe record keyboard sound
Concept Art	

2.10.2.2 Excavation Attempt

Excavation Attempt	
Description	A short brushing sound when an excavation attempt begins
Format	OGG, mono, about a second
Comments	Plays when the player selects a new valid artifact
Concept Art	

2.10.2.3 Fragment Found

Fragment Found	
Description	A positive sounding chime played when an excavation attempt is successful
Format	OGG, mono, about a second
Comments	
Concept Art	

2.10.2.4 Excavation Failure

Excavation Failure	
Description	A soft coin clanking sound to indicate that the player got money from the excavation instead of a fragment
Format	OGG, mono, about a second
Comments	
Concept Art	

2.10.2.5 Fragment Cleaning

Fragment Cleaning	
Description	A gentle polishing sound played while the player cleans a fragment
Format	OGG, mono, short looping sound
Comments	Stops when cleaning ends
Concept Art	

2.10.2.6 Upgrade / Tool Purchase

Upgrade / Tool Purchase	
Description	A purchase confirmation sound played when the player buys an upgrade or tool
Format	OGG, mono, about a second
Comments	
Concept Art	

2.10.2.7 Artifact Assembly

Artifact Assembly	
Description	A layered ceramic sound that plays as the fragments rise, rotate and snap together
Format	OGG, mono, 3-4 seconds
Comments	Maybe add some extra sounds to the end once the artifact finished
Concept Art	

2.10.2.8 Exhibit Completion

Exhibit Completion	
Description	A longer chime or fanfare that plays when an exhibit is completed
Format	OGG, mono, 3-4 seconds
Comments	Should make the player feel like they are progressing well
Concept Art	

3.0 Task list

Week	Team Tasks	Programming Tasks	Art/Audio Tasks	Testing TODO
Week00	• Finish and submit the Game Design Document • Create project schedule • Learn basic Godot workflow and node structure	• Create the Godot project and organize folders • Configure game window and input settings • Create small test scenes to practice Buttons, Control nodes, scene changes, timers, and AnimationPlayer • Experiment with the museum/excavation concept in Godot	• Create initial concept art for museum exhibits and excavation map • Experiment with pixel-art style in Pixelorama • Plan artifact, pedestal, fragment-case, and UI asset sizes	N/A
Week01	• Review project scope and update tasks if needed	• Create Main Menu, Settings, Pause, Exhibit 1, Exhibit 2, Excavation Menu, and Win scenes • Implement scene navigation and navigation arrows • Create HUD structure for funds, reputation, excavation timer, and pause button • Create basic game-state variables for funds, reputation, artifacts, and unlock requirements	• Main Menu background and title splash • Exhibit 1 background • Excavation Map background • Basic UI panels, buttons, and HUD icons • Placeholder artifact, pedestal, and fragment-case sprites	• Test every scene transition • Test menu buttons and pause/settings navigation • Check HUD readability at the game's intended resolution
Week02	• Review core loop and adjust scope if needed	• Implement archaeological site markers and artifact-selection pop-ups • Implement excavation timer, success chance, fragment rewards, and failed-attempt funds rewards • Implement funds and reputation display updates • Add cleaning interaction and fragment case states • Implement Exhibit 1's three artifact displays	• Finalize Exhibit 1 backgrounds and display props • Create fragment-case sprites and pedestal sprites • Create Site Marker sprite states: available, locked, completed • Create initial excavation and fragment sprites • Create UI click, excavation, success, and failure sound effects	• Test excavation success/failure calculations • Test funds rewards after failed attempts • Test collecting and cleaning fragments • Test Exhibit 1 artifact completion flow
Week03	• Review progression pacing	• Implement Exhibit 2 and its locked overlay • Add Exhibit 2 reputation requirement	• Exhibit 2 background and display props • Tool icons and locked-artifact icons	• Test locked Exhibit 2 behavior

		• Implement excavation tool purchases and tool-gated artifacts • Add speed and relic-chance upgrades • Implement passive funds from incomplete and completed artifacts • Connect all win-condition checks	• Lighting overlay sprites • Create remaining artifact sprites • Create cleaning, tool-purchase, and completion sound effects • Create looping museum background music	• Test tool requirements and purchases • Test upgrades and passive-income values • Test progression from Exhibit 1 to Exhibit 2
Week04	• Final GDD revision • Prepare final game build and submission materials	• Create and connect Artifact Assembly Animation using AnimationPlayer • Finish Win Scene and final statistics display • Fix bugs and replace remaining placeholder content • Polish UI feedback, notifications, and scene transitions	• Finalize artifact assembly sprite sheet • Add uncleaned idle, cleaning, assembly, and completed-display animation sequences • Polish UI, background, pedestal, case, and lighting assets • Balance music and sound-effect volume	• Full playthrough from beginning to Win Scene • Balance excavation time, success chances, prices, and requirements • Test every button, pop-up, upgrade, tool, locked state, and scene • Check for visual layering, text readability, audio, and animation issues